PAPER REVIEW

ON

"WILLINGNESS TO PAY FOR BIODIVERSITY CONSERVATION IN DACHIGAM NATIONAL PARK, INDIA"

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1. Introduction

The paper is an attempt at identifying the ***Willingness to Pay (WTP) for the conservation of biodiversity at the Dachigam National Park (DNP), India***. The paper has been written by the joint efforts of ***Mohammad Younus Bhat*** (Department of Economics and International Business, School of Business, University of Petroleum and Energy Studies, Dehradun, 248007, India) and ***Arfat Ahmad Sofi*** (Department of Economics and Finance, Birla Institute of Technology and Sciences (BITS) Pilani, K. K. Birla Goa Campus, Goa - 403726, India). Published in the ***Journal for Nature Conservation*** [Issue no. 62 (2021) 126022], Elsevier.

Among environmental resources, experts place a high priority on biodiversity, especially the contribution of biological resources to human well-being. A broad variety of valuation approaches have been used to value various attributes of biodiversity. These include contingent valuation method to elicit WTP for biodiversity conservation in national parks, willingness to pay for conservation of ecosystem attributes, to examine factors determining WTP, WTP to avoid extinction of species and efficient management of biodiversity in national parks

Biodiversity, encompassing the variety of life across species, ecosystems, and genetic levels, plays a critical role in maintaining ecological integrity and supporting human livelihoods. Biodiversity loss, driven by factors such as fragmentation, destruction, climate change, and pollution, threatens the sustainability of ecosystems globally. Being home to a rich biodiversity, India faces significant challenges in balancing economic development with environmental conservation, especially within its protected areas. From endangerment to extinction of flora and fauna is a wake-up call to all, the economics influenced world thus requires the valuation all environmental resources as a priority.

Economic valuation of biodiversity and ecosystem services has gained importance as a method for translating environmental values into measurable economic terms. By assigning a monetary value to biodiversity, economic valuation supports decision-making that accounts for both the ecological and socio-economic benefits provided by natural resources. For instance, the economic valuation can help prioritize conservation funding, encourage community support, and inform policy design that aligns with both conservation and development goals.

Valuation of environmental and natural resources holds significance in influencing policy and regulations in addition to providing monetary values to them. This way, the environmental sabotage disguised as development projects can be regulated and their impacts, minimized. Correct valuation generates the scope of higher prioritization of conservation and the right compensation, recognizing all the ecosystem services and the resources being affected.

Dachigam National Park is centrally located in Kashmir, 18 km northeast of the capital city of Jammu and Kashmir, Srinagar. The National Park has a varied ecosystem and high diversity with altitudes ranging from 1,700-4,000 m and including the lower and upper Dachigam regions. This altitudinal variation supports a rich variety of flora and fauna, some of the last viable populations of ***Hangul deer*** and ***Asiatic Black Bear***. This makes it more accessible near Srinagar and thus one of the prime locations to be considered for conservation efforts and ecotourism.

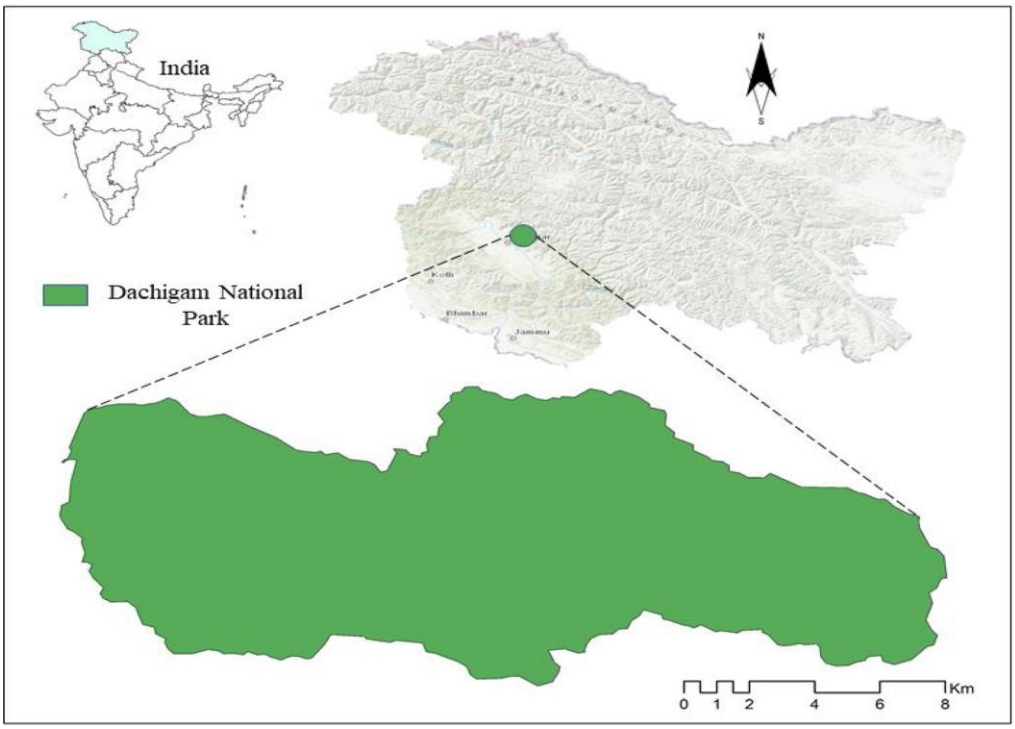

Fig. 1. Location Map of DNP (Source: Prepared by authors using ArcGIS 9.3.)

Dachigam National Park is heavily used, both onsite and offsite, managed at extremely elevated levels of visitor use. Substantial recreational and tourism revenue are generated and contribute to the ecosystem with essential services like water purifying, soil conservation, landscape stability, all which support adjacent communities and agriculture. Thus, the regional economy of livelihood and sustainable development depends on these ecosystem services.

The park has high ecotourism potential with its beautiful landscapes, varied wildlife, and a rich cultural heritage. In the past, it had been a hunting reserve, and there are trekking trails and scopes for bird watching with a possibility of jeep safaris especially in the lower Dachigam area that has approachable roads. Hangul deer is one of the main attractions to both the wildlife enthusiasts and researchers. Seasonal activities, such as trout fishing, enhance the tourism appeal and make DNP a unique destination for eco-friendly tourism initiatives that could foster both conservation and community engagement.

Its climatic condition is ***Sub-mediterranean*** that shows significant variations of temperatures and precipitation. Precipitation is chiefly through ***snowfall***. Rainfall ranges between ***32-546 mm/year***. The summers are mild with maximums reaching 27.3 C, whereas the winters can be harsh with a minimum of around 2.0° C. Dry seasons comprise April to June and the period from September to November both have seasonal effects on flora and water in the park.

Dachigam National Park is blessed with perennial water resources in the form of streams, nallahs, springs, and glaciers. ***Marshar Lake*** is considered the prime source of the Dachigam nallah, one of its principal catchments, which sustains the famous Dal Lake as well.

The National Park is primarily a ***Himalayan moist temperate forest*** and categorized into ***Sub-Alpine*** and ***Alpine forests***. Moist temperate deciduous forests, blue pine forests, and scrub forests have been recorded in the area as well. Alpine grasslands and medicinal herbs are at higher elevations. The lower Dachigam has Prunus, Berberis, and Indigofera species that dominate the shrubs.

Dachigam has excellent varieties of wildlife. Among the most notable are the ***Kashmir Red Deer*** or ***Hangul.*** Other important species are Himalayan Brown Bear, Asiatic Black Bear, Musk Deer, and a variety of bird species. Himalayan Grey Langur, Red Fox, Leopard, and many other smaller mammals such as Himalayan Marten and Indian Porcupine can be seen in the park.

About ***forty-one villages*** share boundaries with the National Park. The villagers depend on agriculture and livestock as their mainstay. Many of them belong to Scheduled Tribe and other economically weaker sections of society, who have limited access to park resources due to the strict forest regulations. ***Human-wildlife conflict*** is a significant issue, as sometimes wildlife from the park damages crops and property, causing tension between locals and park authorities. Other challenges to the park include ***poaching***, ***illegal grazing***, ***encroachment***, and ***limited funding*** for conservation efforts.

Protected areas such as Dachigam National Park face unique challenges in reconciling conservation efforts with the needs of local communities who rely on forest resources for their livelihoods. Local support is essential for successful conservation, as local communities often possess traditional knowledge and practices that contribute to sustainable resource management. Understanding the willingness of these communities to support conservation financially can indicate the level of community engagement and inform strategies to enhance this support.

2. Discussions

The discussions in the document delve into various aspects of the economic valuation of biodiversity conservation in Dachigam National Park (DNP).

The authors estimate the economic value of non-market assets, namely biodiversity, using the generally used survey-based ***Contingent Valuation Method (CVM)***. In this study, the CVM survey compiles main data by means of DNP area household interviews. Out of the 300 households found,279 were selected for the survey. Using a standardized questionnaire, it sought to evaluate their WTP to assist environmental projects. This questionnaire consists of hypothetical situations for better park management covering elements such habitat restoration, preservation of threatened species, and upgraded recreational facilities.

This study used ***Logit*** and ***Probit regression models*** to precisely estimate the WTP by analysing how different socioeconomic and demographic variables affect respondent's choices to make financial contributions. WTP's ***dichotomous (yes/no) response style*** fits conventional CVM guidelines and enables simple factor analysis of those influencing willingness to pay. This modelling clarifies variables influencing WTP behaviour and offers a foundation for knowledge of which groups are more or less likely to encourage financial conservation.

The ***Logit model*** is used to calculate the probability of a specific outcome (here, whether a person is willing to pay for conservation or not). In this context, each participant's response (either "***yes***" or "***no***" to paying) is treated as a dependent variable. The responses are then analysed against a range of factors (independent variables) like income, education, age, and distance from the park, to see how each one affects the likelihood of a "yes" response.

From what we understood, the Logit model tells us how likely someone is to say "yes" or "no based-on factors of influence.

The ***Probit model*** is like the Logit model but uses a slightly different statistical approach. It is used here to calculate the ***probability of willingness to pay***, but it is particularly useful for estimating the ***Mean WTP***.

The Probit model helps show that as the ***suggested payment amount (bid)*** goes up, people are less likely to say "yes." This makes sense because, as the price increases, fewer people find it affordable or worth the cost.

The two approaches let the researchers forecast, depending on age, wealth, etc., which groups of people are more or less likely to back biodiversity conservation. This facilitates the construction of better policies aiming at the most supporting groups and handling the issues of less supportive ones.

The respondents were questioned about their willingness to pay a range of Rs. 20 to Rs. 1000. This setup allows the study to estimate willingness to pay (WTP) for biodiversity preservation under realistic conditions. Those willing to pay have slightly higher average family incomes and slightly larger families. Age and education differences are minor, but there is a slight trend toward younger and more educated respondents being more willing to pay. These trends provide initial insights into how socio-economic factors may influence WTP, suggesting that people with higher economic means may prioritize biodiversity conservation.

The document used several key equations to analyse the data.

$$Zi = \beta 0 + \beta 1X1i + \beta 2X2i + \beta 3X3i + \ldots + \beta nXni + \varepsilon$$

This equation helps predict if a person will say "yes" to paying, depending on their income, age, education, and other factors.

$$Prob(Y = 1Bid) = \beta 0 + \beta 1\ Bid + \varepsilon$$

This equation shows that as the bid amount goes up, people's willingness to pay usually goes down.

$$Mean\ WTP = -(\beta 0 / \beta 1)$$

This equation finds a realistic average payment amount that people are willing to support.

$$WTP_{av} = Mean\ WTP * Total\ Households$$

This equation helps project the total funds that could be raised for park conservation based on everyone's average WTP.

Around ***73%*** of respondents indicated a willingness to pay for improved park management, showing strong community support for conservation. The average Willingness to Pay was calculated as ***Rs 245.57*** which when scaled to the population and area comes out as ***Rs. 289, 772, 600*** which is a substantial potential funding source.

The WTP shows a ***positive correlation*** with a ***greater monthly family income***, implying that financially better-off homes are more likely to support conservation initiatives. Higher WTP comes from respondents with more education levels showing a better awareness of the loss of biodiversity and the advantages of the park.

Proximity to DNP is a key factor. Those that live near the park show a higher WTP since they direct engage with the environment of the park and feel more affected on their daily life. Less ready to pay are older responders and those who are happy with the present management. This trend could result from a reduced sense of the urgency or need for improvements in park management.

The ***Marginal Impacts*** of every variable offer more subtlety. For instance, every one-year rise in age correlates to a 0.6% drop in the possibility of paying; conversely, each extra level of education—based on years of schooling—increases WTP probability by about 0.54%.

The reasons for respondents' reluctance to pay for biodiversity conservation include the belief that conservation is the ***government's responsibility (57.3%)***, ***lack of trust in the proposed management fund (32%)***, and ***satisfaction with current park conditions (6.7%)***.

These results suggest that younger, wealthier, and more educated respondents are more willing to pay, while those satisfied with the current management or who are older are less likely to contribute. The findings underscore the importance of targeting younger and more educated individuals in future awareness campaigns and potentially addressing satisfaction levels with park management to increase WTP.

The paper also suggests that ***lack of awareness*** about biodiversity's broader impact and the benefits of conservation initiatives could contribute to this reluctance. With an eye on raising community support for park management, the study suggests focused educational activities to close this disparity.

Investments in park infrastructure, including establishing separate areas for visitors and environmentally friendly facilities (eco-tourism), could help to improve the management of Dachigam National Park and support sustainable conservation activities. Safaris could also be made more easily available in allowed areas so increasing the financial sustainability of the park.

Potential strategies for improving Dachigam National Park's management and fostering sustainable conservation efforts must include investments in park infrastructure, such as creating separate zones for tourists and eco-friendly amenities (eco-tourism), could improve visitor experience. Safaris could also be made more accessible in permitted zones to make the park more financially sustainable.

The importance of emphasizing the protection of endangered species, like the Hangul deer, and increasing research opportunities within the park should not be ignored as it could improve the populations of biodiversity present.

Another strategy should be the educating of residents about the value of biodiversity and the economic benefits of conservation could increase WTP and overall community involvement in conservation initiatives.

The study concludes that the economic valuation of biodiversity, through approaches like CVM, provides a valuable basis for ***data-driven***, ***community-supported policymaking***. By understanding the residents' preferences and their financial commitment, policymakers can design management strategies that align with both ecological goals and community interests.

To sum it up, the discussion emphasizes that conservation efforts in DNP should be multi-faceted, integrating local economic considerations, targeted awareness programs, and enhanced park infrastructure to foster sustainable management and protect the park's biodiversity for future generations.

Our understanding of the paper given to us is that Willingness to Pay is significant in providing economic value to the natural resources and the environment to bring more light to the importance of their existence and reduce their over-exploitation. One improvement that we think could have been used in the study is the use of ***Travel-cost method*** to better generalize the value for a larger population and have national attention.

3. Conclusion

The study on ***Willingness to pay (WTP) for biodiversity conservation in Dachigam National Park (DNP)*** demonstrates the critical role of economic valuation in understanding community support for conservation efforts. With a huge portion of respondents expressing a willingness to pay to support DNP, the research underscores the local community's recognition of the park's ecological and socio-economic importance.

The analysis reveals that factors such as ***household income***, ***education***, ***proximity to the park***, and ***perceptions of current management*** influence WTP, indicating that these aspects should be central to policy design and community engagement strategies.

The findings suggest that for conservation efforts to be sustainable, policies should not only focus on ecological protection but also prioritize community needs and promote local involvement in park management. Recommendations include educational initiatives to raise awareness about the park's biodiversity, transparent fund management to enhance community trust, and infrastructure development to balance conservation with responsible tourism.

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